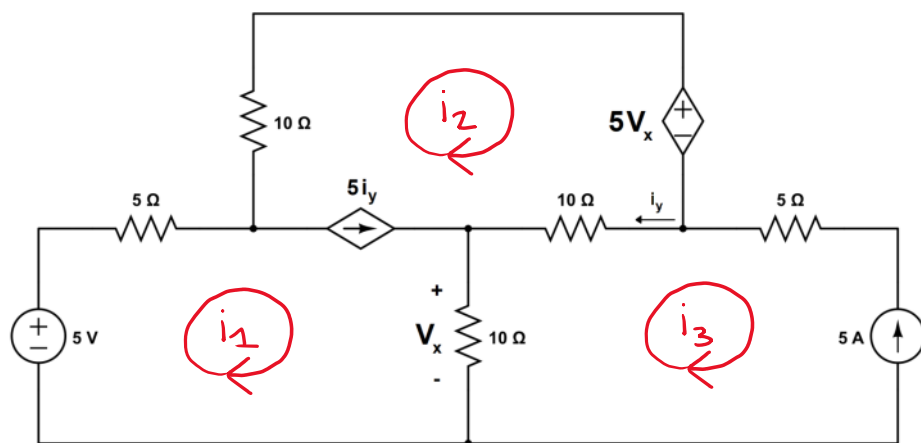


## Practice Quiz 5 Solution

Sunday, October 5, 2025 7:03 PM

For the circuit shown below, use mesh analysis to generate a system of equations for the circuit. You do not have to solve the generated system of equations.



$$i_3 = -5A$$

Supermesh  $i_1 + i_2$ :

$$-5 + 5i_1 + 10(i_1 - i_3) + 10i_2 + 5V_x + 10(i_2 - i_3) = 0$$

Supermesh Constraint Equation:

$$5i_y = i_1 - i_2$$

Dependent Source Constraints:

$$i_y = i_2 - i_3$$

$$V_x = 10(i_1 - i_3)$$

Constraint Simplified:

Supermesh Simplified:

$$5(i_2 - i_3) = i_1 - i_2$$

$$65i_1 + 20i_2 = -245$$

$$i_1 - 6i_2 = 25$$